

(1001CJA100121005)



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

Test Pattern

JEE(Advanced)

REVIEW TEST

14-03-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), NURTURE LIVE PHASE-I & I(B)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	C	B	D	A,B,C,D	A,B,D	A,B,D	A,B,D
SECTION-II	Q.	11	12	3	4	5	6				
	A.	A,B,C	A,B								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	6.00	2.00	8.00	2.00	5.00	10.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	C	D	B	B,D	A,C,D	A,B,C,D	A,B
SECTION-II	Q.	11	12	3	4	5	6				
	A.	A,D	B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.00	2.00	0.60	1.50	4.84	10.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	D	C	D	A,D	A,C	A,C	A,B,C,D
SECTION-II	Q.	11	12	3	4	5	6				
	A.	B,C	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	4.00	2.21 or 2.22	56.00	2.00	4.00	0.50				

(1001CJA100121006)



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

Test Pattern

JEE(Advanced)

REVIEW TEST

14-03-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), NURTURE LIVE PHASE-I & I(B)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B,C,D	A,B	A,B,C	A,B,D	A,B,C,D	B,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	75.00	2.50	1.50	3.60 to 3.61	6.00	34.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	6	8	5	1	2	6				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,D	A,D	A,B,C,D	D	A,B	B,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.25	2.00	13.00	9.00	5.00	0.40				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	4	4	7	4	5				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	B,C	D	A,B	A,B,D	A,C	A,B,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	8.33 or 8.34	1.00	0.50	4.50	3.50	300.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	3	4	6	7	1	5				



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

REVIEW TEST

14-03-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), NURTURE LIVE PHASE-I & I(B)

PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A)

Sol. $[\text{Stress}] = \left[\frac{\text{Force}}{\text{Area}} \right] = [\text{ML}^{-1}\text{T}^{-2}]$

$$[\text{Torque}] = [\vec{r} \times \vec{F}] = [\text{ML}^2\text{T}^{-2}]$$

$$[\text{Work}] = [\vec{F} \cdot \vec{s}] = [\text{ML}^2\text{T}^{-2}]$$

$$[\text{Energy}] = \left[\frac{1}{2}mv^2 \right] = [\text{ML}^2\text{T}^{-2}]$$

2. Ans. (A)

Sol. $[Y] = [V]^x [A]^y [F]^z$
 $\Rightarrow [\text{ML}^{-1}\text{T}^{-2}] = [\text{LT}^{-1}]^x [\text{LT}^{-2}]^y [\text{MLT}^{-2}]^z$
 $\Rightarrow [\text{ML}^{-1}\text{T}^{-2}] = [\text{L}]^{x+y+z} [\text{T}]^{-x-2y-2z} [\text{M}]^z$

Comparing exponents :

$$z = 1$$

$$x + y + z = -1 \Rightarrow x + y = -2 \dots (i)$$

$$\text{and } -x - 2y - 2z \Rightarrow -x - 2y - 2 = -2 \dots (ii)$$

$$\Rightarrow x + 2y = 0 \dots (ii)$$

Solving (i) and (ii)

$$x = -4 \text{ and } y = 2$$

$$\therefore [Y] = [\text{FA}^2\text{V}^{-4}]$$

3. Ans. (A)

Sol. new S.I.

$$n_1 u_1 = n_2 u_2$$

$$n_1 (M_1 L_1^2 T^{-2}) = n_2 (M_2 L_2^2 T^{-2})$$

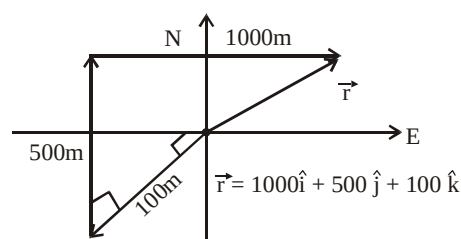
$$n_1 = 1 \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2}$$

$$= 1 \left(\frac{15\text{kg}}{\text{kg}} \right) \left(\frac{20\text{m}}{\text{m}} \right)^2 \left(\frac{20\text{sec}}{\text{sec}} \right)^{-2} = 15$$

4. Ans. (C)

Sol. $\vec{R} = \vec{P} + \vec{Q}$

$$\vec{S} = \vec{Q} - \vec{P}$$



5. Ans. (B)

Sol. The component along z-x plane
 $= \sqrt{z^2 + x^2} = \sqrt{1^2 + 2^2} = \sqrt{5}$

6. Ans. (D)

Sol. $\vec{v} = \frac{\vec{AB}}{|\vec{AB}|} \times v$

7. Ans. (A,B,C,D)

Sol. $[p] = [qr]$, $[s] = [t]$, $[p - qr] = [s + t]$

8. Ans. (A,B,D)

Sol. (A) Only relation which are in multiplication like $(x \propto y^a z^b P^c)$ can be derived.

(B) $V = A^x H^y$

$$L^3 = L^{2x+y}$$

$$\Rightarrow 2x + y = 3$$

No Specific value of x and y

(C) $F = [M]^a [LT^{-2}]^b$

$$[MLT^{-2}] = [M^a L^b T^{-2b}]$$

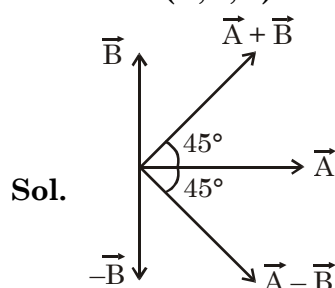
$$a = 1; b = 1$$

Can be derived

(D) $F = \eta A \frac{\Delta V}{\Delta x}$

F which has dimensions of M, L and T only.
 η , A, ΔV and Δx also have dimensions in M, L and T only so three equations and four unknowns can't be solved.

9. Ans. (A,B,D)



Sol.

$$\vec{A} \cdot \vec{B} = AB \cos 90^\circ = 0$$

$$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos 90^\circ} = \sqrt{A^2 + B^2}$$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos 90^\circ} = \sqrt{A^2 + B^2}$$

$$\text{So } |\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$$

$\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$ both are perpendicular

10. Ans. (A,B,D)

Sol. $\vec{A} \cdot \vec{B} = 2 - 4 + 2 = 0$

$$|\vec{A}| = \sqrt{1^2 + 2^2 + 2^2} = 3$$

$$|\vec{B}| = \sqrt{2^2 + 2^2 + 1^2} = 3$$

$$\vec{A} + \vec{B} = 3\hat{i} + 3\hat{k}$$

$$\vec{A} - \vec{B} = -\hat{i} + 4\hat{j} + \hat{k}$$

$$(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = -3 + 0 + 3 = 0$$

11. Ans. (A, B, C)

Sol. Projection on x-y plane will have only x & y component.

12. Ans. (A,B)

Sol. $\vec{a} = 2\hat{i}$

$$\vec{b} = -2\hat{i} + 2\sqrt{3}\hat{j}$$

$$(A) \vec{a} + \vec{b} = 2\sqrt{3}\hat{j}$$

$$(B) \vec{a} - \vec{b} = 4\hat{i} - 2\sqrt{3}\hat{j}$$

$$(C) \vec{a} \cdot \vec{b} = -4$$

$$(D) \vec{b} \cdot \vec{a} = (-2\hat{i} + 2\sqrt{3}\hat{j}) \cdot \hat{i} = -2$$

SECTION-II

1. Ans. 6.00

Sol. $[ML^2T^{-3}] = [LT^{-1}]^a [ML^{-1}]^b [T^{-1}]^c [L]^c$
 $a = 1, b = 2, c = 2$

2. Ans. 2.00

Sol. $R^2 = P^2 + Q^2 + 2PQ \cos \theta$
 $(5Q)^2 = P^2 + Q^2 + 2PQ \cos \theta$
 $Q^2 = P^2 + Q^2 - 2PQ \cos \theta$

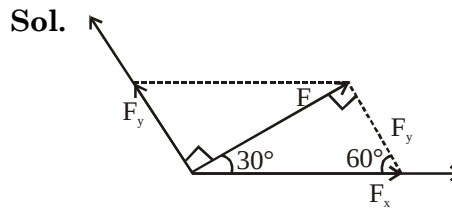
$$\frac{P}{Q\sqrt{3}} = 2$$

3. Ans. 8.00

Sol. $\vec{A} \cdot \vec{B} = 0 \Rightarrow 8 + 24 - 4x = 0$

$$x = \frac{32}{4} = 8$$

4. Ans. 2.00



$$\frac{F}{\sin 60^\circ} = \frac{F_x}{\sin 90^\circ} = \frac{F_y}{\sin 30^\circ}$$

$$\frac{F_x}{F_y} = \frac{\sin 90^\circ}{\sin 30^\circ} = 2$$

5. Ans. 5.00

Sol. $\vec{A} = 2\hat{i} + 3\hat{j}$; Let $\vec{B} = y\hat{i} + z\hat{k}$

$$\vec{A} \cdot \vec{B} = 3y$$

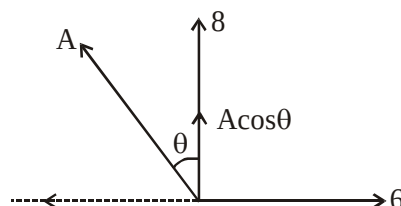
$$15 = 3y$$

$$y = 5$$

$$\vec{B} = 5\hat{i} + z\hat{k}$$

Component of $\vec{B}$ in x-y plane is 5

6. Ans. 10.00



Sol.

$$A \cos \theta = 8 \quad \dots (1)$$

$$A \sin \theta = 6 \quad \dots (2)$$

$$(1)^2 + (2)^2$$

$$A^2 (\cos^2 \theta + \sin^2 \theta) = 8^2 + 6^2$$

$$A^2 = 100$$

$$A = 10$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(D)

2. Ans.(B)

3. Ans.(D)

4. Ans.(C)

5. Ans.(D)

6. Ans.(B)

7. Ans.(B,D)

8. Ans.(A,C,D)

9. Ans.(A,B,C,D)

10. Ans.(A,B)

11. Ans.(A,D)

12. Ans.(B,D)

SECTION-II

1. Ans.(3.00)

2. Ans.(2.00)

3. Ans.(0.60)

4. Ans.(1.50)

5. Ans.(4.84)

6. Ans.(10.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

Sol. $\frac{+}{-8/3} \text{---} \frac{-}{21/4} \frac{+}{+}$
 $\Rightarrow x \in \left[\frac{-8}{3}, \frac{21}{4} \right]$

2. **Ans. (C)**

Sol. Use AM $\geq$ GM

$$\frac{\frac{x}{3} + \frac{27}{x}}{2} \geq \sqrt{\frac{x}{3} \cdot \frac{27}{x}}$$

$$\Rightarrow \frac{x}{3} + \frac{27}{x} \geq 6$$

3. **Ans. (B)**

Sol. $2 \times 64 - 16 - 22 \times 4 + \lambda = 0$

$$\Rightarrow \lambda = -24$$

Now divide $2x^3 - x^2 - 22x - 24$ with $x - 4$ we

get other roots as -2 and $\frac{-3}{2}$

4. **Ans. (D)**

Sol. Do yourself

5. **Ans. (C)**

Sol. $\frac{3\sqrt{2}(\sqrt{6}-\sqrt{3})}{6-3} - \frac{4\sqrt{3}(\sqrt{6}-\sqrt{2})}{6-2} + \frac{\sqrt{6}(\sqrt{3}-\sqrt{2})}{3-2} + \frac{\sqrt{5-\sqrt{21}}}{\sqrt{25-21}}$
 $= \sqrt{2}(\sqrt{6}-\sqrt{3}) - \sqrt{3}(\sqrt{6}-\sqrt{2}) + \sqrt{6}(\sqrt{3}-\sqrt{2}) + \frac{\sqrt{10-2\sqrt{21}}}{2\sqrt{2}}$
 $= \sqrt{12} - \sqrt{6} - \sqrt{18} + \sqrt{6} + \sqrt{18} - \sqrt{12} + \frac{\sqrt{(\sqrt{7}-\sqrt{3})^2}}{2\sqrt{2}}$
 $= \frac{\sqrt{7}-\sqrt{3}}{\sqrt{8}} = \sqrt{\frac{7}{8}} - \sqrt{\frac{3}{8}}$

$$\therefore a = 7, b = 3 \Rightarrow a + b = 10$$

6. **Ans. (D)**

Sol. $\therefore (x-3)(x+2) \leq 0$

$$\frac{+}{-2} \text{---} \frac{-}{3} \frac{+}{+}$$

$$x \in [-2, 3]$$

$\therefore$ Natural values of $x = 1, 2, 3$

so, $A = \{1, 2, 3\}$

$$\therefore n(A) = 3$$

$$\text{Number of proper subset} = 2^3 - 1 = 7$$

7. **Ans. (A,D)**

Sol. $\left[\frac{x^{\frac{2}{3}+\frac{2}{3}}}{y^{\frac{1}{1}+\frac{1}{2}}} \times \left(\frac{x^{1+1}}{y^{1+2}} \right)^{\frac{1}{2}} \right]^9 = \left[\frac{x^{\frac{4}{3}+1}}{y^{\frac{3}{2}+\frac{3}{2}}} \right]^9$
 $= \left(\frac{x^{\frac{7}{3}}}{y^3} \right)^9 = \frac{x^{21}}{y^{27}}$

8. **Ans. (A,C)**

Sol. $P(x) = (x+1)^2$ has roots 0, 1, 2

$$\Rightarrow P(x) - (x+1)^2 = 2x(x-1)(x-2)$$

$$\Rightarrow P(x) = 2x(x-1)(x-2) + (x+1)^2$$

$$P(-1) = 2 \times (-1) \times (-2) \times (-3) = -12$$

$$P(3) = 2 \times 3 \times 2 \times 1 + 16 = 28$$

9. **Ans. (A,C)**

Sol. Let $(\sqrt{3+2\sqrt{2}})^{3x} = t$

$$\therefore t + \frac{1}{t} = 2\sqrt{2}$$

$$\Rightarrow t^2 + \frac{1}{t^2} + 2 = 8$$

$$\Rightarrow (t^2)^2 - 6(t^2) + 1 = 0$$

$$\Rightarrow t^2 = 3 \pm 2\sqrt{2}$$

$$\Rightarrow (3+2\sqrt{2})^{3x} = (3+2\sqrt{2})^1 \text{ or } (3-2\sqrt{2})^{-1}$$

$$\Rightarrow 3x = 1 \text{ or } -1$$

$$\Rightarrow \boxed{x = \frac{1}{3} \text{ or } \frac{-1}{3}}$$

10. **Ans. (A,B,C,D)**

Sol. $t = x + \frac{1}{x}$

$$\Rightarrow t^2 - 10t + 21 = 0$$

$$\Rightarrow t = 7, 3$$

$$\frac{x^2+1}{x} = 7 \quad \Rightarrow \quad \frac{x^2+1}{x} = 3$$

$$x^2 - 7x + 1 = 0$$

$$x^2 - 3x + 1 = 0$$

$$x = \frac{7 \pm \sqrt{45}}{2}$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

11. Ans. (B,C)

Sol. $(x+1)(x+4)(x^2+4x+4) = 2x^2$
 $(x^2+5x+4)(x^2+4x+4) = 2x^2$

$$\Rightarrow \left(x+5+\frac{4}{x}\right)\left(x+4+\frac{4}{x}\right) = 2$$

Let $x+4+\frac{4}{x} = t$

$$(t+1)t = 2 \Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t = 1, -2$$

Now $x+4+\frac{4}{x} = 1$ or $x+4+\frac{4}{x} = -2$

$$\Rightarrow x^2 + 3x + 4 = 0 \quad x^2 + 6x + 4 = 0$$

Sum of roots = -3 Sum of roots = -6

$$\Rightarrow -3 + -6 = -9$$

$$-9 > -10 \text{ \& } -9 > -11$$

12. Ans. (A,C)

Sol. $\frac{p^2}{q^2} = \frac{(x+1)^2}{(x-1)^2}$

$$\Rightarrow \frac{p^2 - q^2}{p^2 + q^2} = \frac{(x+1)^2 - (x-1)^2}{(x+1)^2 + (x-1)^2} = \frac{2 \cdot (2x)}{2(x^2 + 1)}$$

$$\Rightarrow f(x) = \frac{(2x)}{(x^2 + 1)} \quad \text{Now check options.}$$

SECTION-II

1. Ans. 4.00

Sol. Let $(x+1)(x+4)(x^2+4x+4) = 2x^2$

$$\Rightarrow \left(x+5+\frac{4}{x}\right)\left(x+4+\frac{4}{x}\right) = 2$$

Let $t = x + \frac{4}{x} + 4$

$$\Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t = 1, -2$$

Now $x+4+\frac{4}{x} = 1$ $\Rightarrow x^2 + 3x + 4 = 0$ $\Rightarrow D < 0$ (no real Sol.)	$x+4+\frac{4}{x} = -2$ $x^2 + 6x + 4 = 0$ $\Rightarrow x = -3 \pm \sqrt{5}$ $x_1 x_2 = -(\sqrt{5}-3)(\sqrt{5}+3)$ $= 4$
---	---

2. Ans. 2.21 or 2.22

Sol. Given expression is

$$(7x-5)^2 + (2y-1)^2 + (3z-1)^2 = 0 \text{ then the}$$

value of $x+y+3z$ is $\frac{5}{7} + \frac{1}{2} + 1 = 2.21 \text{ or } 2.22$

3. Ans. 56.00

Sol. $xy = \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right) = 1$

$$x^2 + y^2 = 4$$

$$x-y = -\sqrt{2}, \quad x+y = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6}$$

$$(x-y) + 2xy(x+y) - xy(x^3 - y^3)$$

$$\Rightarrow (x-y) + 2xy(x+y) - xy(x-y)(x^2 + y^2 + xy)$$

$$= -\sqrt{2} + 2 \times 1 (\sqrt{6}) + \sqrt{2} (4+1)$$

$$= -\sqrt{2} + 2(\sqrt{6}) + 5\sqrt{2}$$

$$= 4\sqrt{2} + 2\sqrt{6}$$

$$= \sqrt{32} + \sqrt{24}$$

4. Ans. 2.00

Sol. $\therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{7}$

$$\Rightarrow 7x + 7y = xy$$

$$\Rightarrow xy - 7x - 7y = 0$$

$$\Rightarrow xy - 7x - 7y + 49 = 49$$

$$\Rightarrow (x-7)(y-7) = 49$$

$x-7$	$y-7$
1	49
49	1
7	7

$$\Rightarrow (x,y) \equiv (8,56) \text{ reject}$$

$$\Rightarrow (x,y) \equiv (56,8)$$

$$\Rightarrow (x,y) \equiv (14,14)$$

5. Ans. 4.00

Sol. $\frac{(5-1)(5+1)(5^2+1)\dots(5^n+1)}{4} + \frac{1}{4} = \frac{5^{n^2+4n-288}}{4}$

$$\Rightarrow \frac{5^{2n} - 1 + 1}{4} = \frac{5^{n^2+4n-288}}{4}$$

$$\Rightarrow 5^{2n} = 5^{n^2+4n-288}$$

$$\Rightarrow n^2 + 2n - 288 = 0$$

$$\Rightarrow n = 16 \text{ or } n = -18 \text{ (reject)}$$

6. Ans. 0.50

Sol. $(x^2)^2 + (2y^2)^2 + 2x^2(2y^2) - 4x^2y^2 + 3x^2y^2$

$$= (x^2 + 2y^2)^2 - (xy)^2$$

$$= (x^2 + 2y^2 - xy)(x^2 + 2y^2 + xy)$$

$$\Rightarrow a = 1, b = 2$$

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A, B, C, D)

Sol. $y = \frac{a}{t^2} e^{-\left(\frac{t-x}{b-c}\right)}$

$$\left[\frac{t}{b}\right] = \left[\frac{x}{c}\right] = M^0 L^0 T^0$$

$$\Rightarrow [b] = T \text{ \& } [c] = [L] \text{ \& }$$

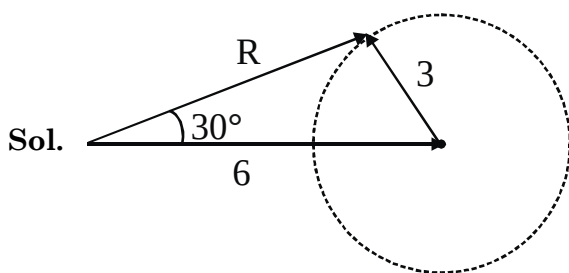
$$\left[\frac{a}{t^2}\right] = y \Rightarrow [a] = [LT^{-2}] \text{ \& }$$

$$e^{-\left(\frac{t-x}{b-c}\right)} \text{ is dimensionless}$$

2. Ans. (A, B)

Sol. Energy can be formulated using mass and velocity. Same for length, acceleration and frequency.

3. Ans. (A,B,C)



4. Ans. (A, B, D)

Sol. Clearly $-\vec{E} = \vec{A} + \vec{B}$

$$\text{or } \vec{E} = -\vec{A} - \vec{B}$$

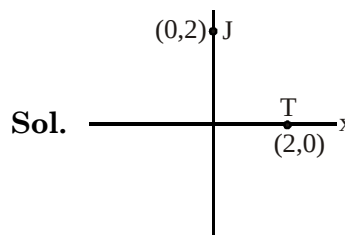
$$\vec{D} - \vec{C} = \vec{D} + (-\vec{C}) = \vec{A}$$

$$\text{Then } \vec{E} + \vec{D} - \vec{C} = \vec{E} + \vec{A} = -\vec{B}$$

5. Ans. (A,B,C,D)

Sol. Vector upon magnitude of vector is unit vector. Hence (A) & (C) is correct. If two vector each of magnitude one at 120° then their resultant has magnitude one. Hence B & D is correct.

6. Ans. (B,D)



$$\vec{r}_{CT} = 3\hat{i} + 3\hat{j}$$

$$\vec{r}_C - \vec{r}_T = 3\hat{i} + 3\hat{j}$$

$$\vec{r}_C = 5\hat{i} + 3\hat{j}$$

$$\vec{r}_{CJ} = \sqrt{5^2 + 1^2} = \sqrt{26}$$

$$\vec{r}_{CT} = \sqrt{3^2 + 3^2} = \sqrt{18}$$

SECTION-II

1. Ans. 75.00

Sol. $\vec{A} = 6\hat{i} + 2\hat{j} + 3\hat{k}$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{6\hat{i} + 2\hat{j} + 3\hat{k}}{7}$$

$$\hat{A} = \frac{6}{7}\hat{i} + \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k}$$

$$\vec{F}_A = 14(\hat{A}) = 12\hat{i} + 4\hat{j} + 6\hat{k} \quad \dots(1)$$

$$\vec{B} = 3\hat{i} - 2\hat{j} + 6\hat{k}$$

$$\hat{B} = \frac{3\hat{i}}{7} - \frac{2\hat{j}}{7} + \frac{6\hat{k}}{7}$$

$$\vec{F}_B = 7\hat{B} = 3\hat{i} - 2\hat{j} + 6\hat{k} \quad \dots(2)$$

$$\vec{C} = 2\hat{i} - 3\hat{j} - 6\hat{k}$$

$$\hat{C} = \frac{2\hat{i}}{7} - \frac{3\hat{j}}{7} - \frac{6\hat{k}}{7}$$

$$\vec{F}_C = 7\hat{C} = 2\hat{i} - 3\hat{j} - 6\hat{k}$$

$$\vec{r}_A = 2\hat{i} - 1\hat{j} - 3\hat{k}$$

$$\vec{r}_B = 5\hat{i} - 1\hat{j} + 1\hat{k}$$

Displacement

$$= \vec{r}_B - \vec{r}_A = 5\hat{i} - 1\hat{j} + 1\hat{k} - 2\hat{i} + 1\hat{j} + 3\hat{k}$$

$$= 3\hat{i} + 4\hat{k}$$

$$F_{\text{Net}} = \vec{F}_A + \vec{F}_B + \vec{F}_C = 17\hat{i} - \hat{j} + 6\hat{k}$$

$$\text{work done} = \vec{F}_{\text{Net}} \cdot (\text{displacement})$$

$$= 51 + 24 = 75 \text{ Joule Ans.}$$

2. Ans. 2.50

$$\text{Sol. } V = [A]^x [d]^y [g]^z$$

$$LT^{-1} = L^{2x} M^y L^{-3y} L^z T^{-2z}$$

$$M^0 L^1 T^{-1} = M^y L^{2x-3y+z} T^{-2z}$$

By solving $x = 1/4$, $y = 0$, $z = 1/2$

$$\frac{v_1}{v_2} = \left(\frac{A_1}{A_2} \right)^{1/4}; \frac{v_1}{v_2} = \left(\frac{L_1}{L_2} \right)^{1/2}; \frac{10}{v} = \left[\frac{1}{10^3} \right]^{1/2}$$

$$v = 10 \times 10^{3/2} = 10^{5/2}$$

3. Ans. 1.50

$$\text{Sol. } [f] = [k_1] \left[\frac{m}{kT^2} \right]^{3/2} [v^2]$$

$$\left[\frac{mv^2}{kT} \right] = 1 \text{ (exp. function are dimensionless)}$$

$$\left[\frac{m}{kT} \right] = \left[\frac{1}{v^2} \right]$$

$$1 = [k_1] \left[\frac{1}{v^2 \theta} \right]^{3/2} [v^2]$$

$$1 = [k_1] \left[\frac{v^2}{v^3 \theta^{3/2}} \right]$$

$$[k_1] = [LT^{-1} \theta^{3/2}]$$

$$a + b + c = \frac{3}{2} = 1.5$$

4. Ans. 3.60 to 3.61

Sol. Displacement = $\overline{AB}$, angle between $\vec{r}_1$ and

$$\vec{r}_2 : \theta = 75^\circ - 15^\circ = 60^\circ$$

$$\text{From figure, } AB^2 = r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta$$

$$= 3^2 + 4^2 - 2 \times 3 \times 4 \cos 60^\circ = 13$$

$$\Rightarrow AB = \sqrt{13}$$

5. Ans. 6.00

$$\text{Sol. } \vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t$$

6. Ans. 34.00

$$\text{Sol. } \vec{r}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{s}_1 = 30 \times \frac{(\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{1^2 + 2^2 + 2^2}}$$

$$\vec{s}_2 = 15 \times \frac{(-\hat{i} - 2\hat{j} - 2\hat{k})}{\sqrt{1^2 + 2^2 + 2^2}}$$

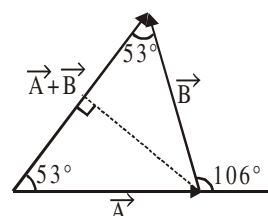
$$\vec{r}_f = \vec{r}_1 + \vec{s}_1 + \vec{s}_2$$

$$= 7\hat{i} + 13\hat{j} + 14\hat{k}$$

$$x + y + z = 7 + 13 + 14 = 34$$

SECTION-III

1. Ans. 6



Sol.

so vector Δ is an isocelse Δ .

$$|\vec{A} + \vec{B}| = |\vec{A}| \cos 53^\circ \times 2 = 5 \times \frac{3}{5} \times 2N = 6N$$

2. Ans. 8

$$\text{Sol. } n_1 [kg \ m^2 s^{-2}] = 1 [2kg \times 16 \ m^2 \times \frac{1}{4} \ sec^{-1}]$$

$$n_1 = 8$$

3. Ans. 5

$$\text{Sol. } [L] = [ML^2 T^{-2}]^a [T]^b [ML^{-3}]^e$$

$$[L] = [ML^{a+c}][L^{2a-3c}][T^{-2a+b}]$$

$$a + c = 0$$

$$2a - 3c = 1$$

$$-2a + b = 0$$

$$2a + 3a = 0$$

$$5a = 1$$

$$a = \frac{1}{5}$$

$$c = -\frac{1}{5}$$

$$b = +\frac{2}{5}$$

$$\Rightarrow x = 5$$

$$y = +5$$

$$x = 5$$

$$\frac{5+5+5}{3} = 5$$

4. Ans. 1

Sol. $P_0 = F^a P^b M^c = M^{a+b+c} L^{a+b} T^{-2a-b} = M^1 L^2 T^{-3}$

$\Rightarrow a + b = 2 \quad \dots(i)$

$\Rightarrow a + b + c = 1 \quad \dots(ii)$

$\Rightarrow 2a + b = 3 \quad \dots(iii)$

$\Rightarrow c = -1 \Rightarrow a = 1 \Rightarrow 6$

$P_0 = FPM^{-1}$

5. Ans. 2

Sol. $\frac{v^n}{rg} = \tan \theta \Rightarrow [v]^n = [r][g]$

$\Rightarrow [LT^{-1}]^n = [L] [LT^{-2}]$

$\Rightarrow L^n T^{-n} = L^2 T^{-2}$

On comparing

$n = 2$

6. Ans. 6

Sol. Magnitude of component of acceleration in the direction of velocity

$\vec{a} \cdot \hat{v} = (6\hat{i} + 2\hat{j} - 6\hat{k}) \cdot \hat{i} = 6 \text{ m/s}^2$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,D)

2. Ans. (A,D)

3. Ans. (A,B,C,D)

4. Ans. (D)

5. Ans. (A,B)

6. Ans. (B,C,D)

SECTION-II

1. Ans. (3.25)

2. Ans. (2.00)

3. Ans. (13.00)

4. Ans. (9.00)

5. Ans. (5.00)

6. Ans. (0.40)

SECTION-III

1. Ans. (4)

2. Ans. (4)

3. Ans. (4)

4. Ans. (7)

5. Ans. (4)

6. Ans. (5)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,C)

Put $x = k$ in $(x^3 + x^2 - k^2x + x - 2) = 0$

$\therefore k^3 + k^2 - k^3 + k - 2 = 0$

$\Rightarrow (k + 2)(k - 1) = 0$

$\Rightarrow k = 1 \text{ or } -2$

2. Ans. (D)

$4 - \sqrt{x} = \sqrt{2x - 68}$

$\Rightarrow 16 - 8\sqrt{x} + x = 2x - 68$

$\Rightarrow x + 8\sqrt{x} - 84 = 0$; let $\sqrt{x} = t$

$\Rightarrow t^2 + 8t - 84 = 0$

$\Rightarrow (t + 14)(t - 6) = 0$

$\Rightarrow t = 6 \text{ or } t = -14$

$\Rightarrow \sqrt{x} = 6 \text{ (reject)} \quad \sqrt{x} = -14 \text{ (reject)}$

for such value of x ,

L.H.S become -ve

and R.H.S become

+ve

3. Ans. (A,B)

$\sqrt{\lambda} = \sqrt{4 + \sqrt{7}} - \sqrt{4 - \sqrt{7}}$

$\Rightarrow \lambda = \left(\sqrt{4 + \sqrt{7}} - \sqrt{4 - \sqrt{7}} \right)^2$

$\Rightarrow \lambda = (4 + \sqrt{7}) + (4 - \sqrt{7}) - 2\sqrt{(4 + \sqrt{7})(4 - \sqrt{7})}$

$\Rightarrow \lambda = 8 - 2\sqrt{16 - 7}$

$\Rightarrow \lambda = 2$

4. Ans. (A,B,D)

$\therefore a^3 + b^3 = 800$

$\Rightarrow (a + b)(a^2 + b^2 - ab) = 800$

$\Rightarrow (a + b)[(a + b)^2 - 3ab] = 800$

$\Rightarrow 20[400 - 3ab] = 800$

$\Rightarrow 400 - 3ab = 40$

$\Rightarrow 3ab = 360$

$\Rightarrow ab = 120$

$\Rightarrow ab = 2^3 \cdot 3^1 \cdot 5^1$

$\therefore x = 3, y = 1, z = 1$ now check options.

5. Ans. (A,C)

Apply A.M. $\geq$ G.M

$\Rightarrow \frac{8a}{b} + \frac{4b}{a} \geq \left(\frac{8a}{b} \cdot \frac{4b}{a} \right)^{1/2}$

$$\Rightarrow \frac{8a}{b} + \frac{4b}{a} \geq 8\sqrt{2}$$

$\therefore k = 8$ Now check options.

6. **Ans. (A,B,D)**

$$\text{Let } a + \frac{1}{a} = k$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3.a.\frac{1}{a}\left(a + \frac{1}{a}\right) = k^3$$

$$\Rightarrow 18 + 3(k) = k^3$$

$$\Rightarrow k^3 - 3k - 18 = 0$$

$$\Rightarrow (k-3)(k^2 + 3k + 6) = 0$$

$$\Rightarrow k = 3 \text{ or } k^3 + 3k + 6 = 0 \text{ No. sol.}$$

SECTION-II

1. **Ans. 8.33 or 8.34**

$$[(x+1)(x+4)][(x+2)(x+3)] = 120$$

$$\Rightarrow [x^2 + 5x + 4][x^2 + 5x + 6] = 120$$

$$\text{Let } x^2 + 5x = t$$

$$\Rightarrow [t+4][t+6] = 120$$

$$\Rightarrow t^2 + 10t - 96 = 0$$

$$\Rightarrow (t+16)(t-6) = 0$$

$$\Rightarrow t-6=0 \quad \text{or} \quad t+16=0$$

$$\Rightarrow x^2 + 5x - 6 = 0 \quad x^2 + 5x + 16 = 0$$

$$\Rightarrow (x+6)(x-1) = 0 \quad \text{No. sol.}$$

$$\Rightarrow x = 1 \text{ or } -6$$

2. **Ans. 1.00**

$$(2)^{2/x} - 3.(2)^{1/x}.(2)^1 + 8 = 0$$

$$\Rightarrow ((2)^{1/x})^2 - 6(2)^{1/x} + 8 = 0$$

$$\text{Let } 2^{1/x} = t$$

$$\Rightarrow t^2 - 6t + 8 = 0$$

$$\Rightarrow t = 2 \text{ or } t = 4$$

$$\Rightarrow (2)^{1/x} = (2)^1 \quad \Rightarrow (2)^{1/x} = (2)^2$$

$$\Rightarrow \frac{1}{x} = 1 \quad \Rightarrow \frac{1}{x} = 2$$

Note : for $\sqrt[n]{a}$; $x \in \mathbb{N} - \{1\}$

$$\Rightarrow x = 1 \text{ (reject)} \quad \Rightarrow x = \frac{1}{2} \text{ (reject)}$$

$\therefore$ Number of solution = 0

3. **Ans. 0.50**

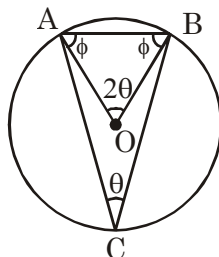
$$\therefore \phi + \phi + 2\theta = 180^\circ$$

$$\Rightarrow \phi + \theta = 90^\circ$$

$$\Rightarrow \phi + 30^\circ = 90^\circ$$

$$\Rightarrow \phi = 60^\circ$$

$$\text{so, } \frac{30}{k} = \frac{30}{60} = 0.50$$



4. **Ans. 4.50**

$$* \quad \frac{f(x)}{x-2} \Rightarrow \text{Remainder} = 3$$

$$\therefore f(2) = 3$$

$$* \quad \frac{f(x)}{x+2} \Rightarrow \text{Remainder} = 5$$

$$\therefore f(-2) = 5$$

$$\therefore f(x) = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$$\Rightarrow f(x) = (x^2 - 4).Q(x) + (px + q)$$

Put $x = 2$	Put $x = -2$
↓	↓
$f(2) = 2p + q$	$f(-2) = -2p + q$
$2p + q = 3 \quad \dots(1)$	$q - 2p = 5 \quad \dots(2)$

$$\text{solve (1) and (2)} \Rightarrow \left. \begin{aligned} p &= -\frac{1}{2} \\ q &= 4 \end{aligned} \right\}$$

$$\therefore q - p = 4 + \frac{1}{2} = 4.50$$

5. **Ans. 3.50**

$$(x^2 + 6)^{7-x} = 1 \quad * \quad 7 - x = 0$$

$$* \quad x^2 + 6 = 1$$

$$x = 7$$

$$\Rightarrow x^2 = -5$$

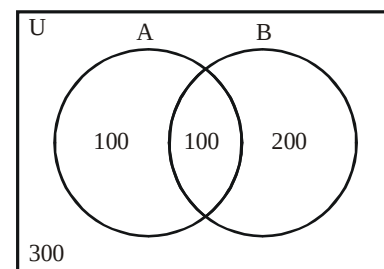
(reject)

$$\therefore \text{sum of values of } x = 7$$

$$\Rightarrow k = 7$$

$$\text{so, } \frac{k}{2} = 3.50$$

6. **Ans. 300.00**



$$\therefore n(A' \cap B') = 300$$

SECTION-III

1. **Ans. 3**

$$\therefore 8x^3 + (-27x^6) + y^3 = -18x^3y$$

$$\Rightarrow (2x)^3 + (-3x^2)^3 + (y)^3 = 3(2x)(-3x^2)(y)$$

{ $\therefore$ We know that, If $a^3 + b^3 + c^3 = 3abc$ then $a + b + c = 0$ or $a = b = c$ }

$$\Rightarrow 2x + (-3x^2) + y = 0 \quad \text{or} \quad 2x = -3x^2 = y$$

(reject)

$$\Rightarrow 2x + y = 3x^2$$

$$\Rightarrow \frac{2}{x} + \frac{y}{x^2} = 3$$

2. Ans. 4

$$\frac{A_1}{A_2} = \frac{\frac{1}{2}r_1^2\theta}{\frac{1}{2}r_2^2\theta} = \frac{r_1^2}{r_2^2} = \frac{(10)^2}{(5)^2} = 4$$

3. Ans. 6

$$\begin{aligned} & \sqrt{(5+\sqrt{6})^2} - \sqrt{(3+\sqrt{2})^2} - \sqrt{(\sqrt{3}-1)^2} - 2 \\ &= (5+\sqrt{6}) - (3+\sqrt{2}) - (\sqrt{3}-1) - 2 \\ &= 5 + \sqrt{6} - 3 - \sqrt{2} - \sqrt{3} + 1 - 2 \\ &= \sqrt{6} - \sqrt{2} - \sqrt{3} + 1 \\ &= \sqrt{2}(\sqrt{3}-1) - (\sqrt{3}-1) \\ &= (\sqrt{3}-1)(\sqrt{2}-1) \end{aligned}$$

$$\therefore \begin{matrix} a = 3 & \text{or} & a = 2 \\ b = 2 & & b = 3 \end{matrix}$$

then $ab = 6$

4. Ans. 7

$$A = \{2, 4, 6, 8, \dots, 100\}$$

$$B = \{3, 6, 9, \dots, 99\}$$

$$C = A \cap B = \{6, 12, 18, \dots, 96\}$$

$$\therefore n(C) = 16$$

$$\text{sum of digits} = 1 + 6 = 7$$

5. Ans. 1

$$3^{3x} - 4 \cdot 3^{2x} + 3 \cdot 3^x = 0$$

$$\Rightarrow 3^x [3^{2x} - 4 \cdot 3^x + 3] = 0$$

$$\{\text{Let } 3^x = t, \Rightarrow 3^{2x} = t^2\}$$

$$\Rightarrow t[t^2 - 4t + 3] = 0$$

$$\Rightarrow t(t-1)(t-3) = 0$$

$$\Rightarrow t = 0, 1, 3$$

$$\begin{array}{l} 3^x = 0 \quad \left| \begin{array}{l} 3^x = 1 \\ 3^x = 3 \end{array} \right. \\ \Rightarrow (\text{reject}) \Rightarrow x = 0 \quad \left| \begin{array}{l} x = 1 \\ (\text{reject}) \end{array} \right. \end{array}$$

6. Ans. 5

$$\therefore \left(\frac{x-4}{x} \right)^{x-2} = \left(\frac{x}{x-4} \right)^{2x-3}$$

$$\Rightarrow \left(\frac{x-4}{x} \right)^{x-2} = \left(\frac{x-4}{x} \right)^{3-2x}$$

$$* \quad \frac{x-4}{x} = 1$$

$$\Rightarrow x - 4 = x$$

no solution.

$$* \quad \frac{x-4}{x} = -1$$

$$\Rightarrow x = 2 \text{ (reject)}$$

$$\text{sum} = \frac{5}{3} = \lambda$$

$$\Rightarrow 3\lambda = 5$$

$$* \quad x - 2 = 3 - 2x$$

$$\Rightarrow x = \frac{5}{3}$$

$$* \quad \frac{x-4}{x} = 0$$

$$x = 4 \text{ reject}$$